Young Tableaux

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1 Introduction

Young tableaux play a central role in the study of group theory, particularly in the calculation of the dimensions of irreducible representations. They provide a combinatorial framework that simplifies many otherwise complex calculations, especially in the context of symmetric and unitary groups. Because of this, Young tableaux are widely used as a powerful tool in both mathematical and physical applications of group theory.

2 Find Dimensionality of Any Tableaux in SU(N)

In the representation theory of the $SU(N)$ group, the dimensionality of a given Young tableau can be obtained by counting the number of ways to fill its boxes with the numbers $1, 2, \ldots, N$ under certain rules. Such fillings are known as *semistandard Young tableaux (SSYT)*. The rules are:

1. Fill the tableau row by row: start at the leftmost box of the first row and proceed to the right; after finishing one row, move to the next.
2. Within each row, the numbers must be **weakly increasing** from left to right (i.e. they do not decrease).
3. Within each column, the numbers must be **strictly increasing** from top to bottom.
4. Continue filling until all boxes are filled following these conditions.
5. The number of distinct valid fillings is the dimension of the irreducible representation of $SU(N)$ corresponding to that tableau shape.

2.1 Examples

Consider the tableau with a single box:

$$\boxed{}$$

- In $SU(2)$: possible fillings are $\boxed{1}, \boxed{2} \Rightarrow$ dimension $= 2$. - In $SU(3)$: possible fillings are $\boxed{1}, \boxed{2}, \boxed{3}$ $\Rightarrow$ dimension $= 3$.

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Consider the tableau with two boxes in a row:

$$\begin{array}{|c|c|}\hline & \\ \hline \end{array}$$

- In $SU(2)$: valid fillings are $\begin{array}{|c|c|}\hline 1 & 1 \\ \hline \end{array}$, $\begin{array}{|c|c|}\hline 1 & 2 \\ \hline \end{array}$, $\begin{array}{|c|c|}\hline 2 & 2 \\ \hline \end{array}$ $\Rightarrow$ dimension $= 3$. - In $SU(3)$: valid fillings are $\begin{array}{|c|c|}\hline 1 & 1 \\ \hline \end{array}$, $\begin{array}{|c|c|}\hline 1 & 2 \\ \hline \end{array}$, $\begin{array}{|c|c|}\hline 1 & 3 \\ \hline \end{array}$, $\begin{array}{|c|c|}\hline 2 & 2 \\ \hline \end{array}$, $\begin{array}{|c|c|}\hline 2 & 3 \\ \hline \end{array}$, $\begin{array}{|c|c|}\hline 3 & 3 \\ \hline \end{array}$ $\Rightarrow$ dimension $= 6$.

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Consider the tableau with two boxes in a column:

$$\begin{array}{|c|}\hline \\ \hline \\ \hline \end{array}$$

- In $SU(2)$: only $\begin{array}{|c|}\hline 1 \\ \hline 2 \\ \hline \end{array}$ is allowed $\Rightarrow$ dimension $= 1$. - In $SU(3)$: valid fillings are $\begin{array}{|c|}\hline 1 \\ \hline 2 \\ \hline \end{array}$, $\begin{array}{|c|}\hline 1 \\ \hline 3 \\ \hline \end{array}$, $\begin{array}{|c|}\hline 2 \\ \hline 3 \\ \hline \end{array}$ $\Rightarrow$ dimension $= 3$.

Note: A column with three boxes $\begin{array}{|c|}\hline \\ \hline \\ \hline \\ \hline \end{array}$ cannot exist in $SU(2)$ because there are not enough distinct numbers, but in $SU(3)$ there is exactly one valid filling $\begin{array}{|c|}\hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array}$.

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This counting method gives the dimension of the tableau. However, there exists a more direct combinatorial formula called the *Hook Length Formula* (or *hook number rule*), which often makes the computation much simpler.

3 Using Hook Number Method

3.1 What is Hook Number ??

In a tableau, for each box a number has been assigned and that represents the hook number value. For calculating the hook number, follow these steps:

1. The box for which you want to calculate the hook number, go to that row and go towards very right of that row (Outside the Tableaux).
2. Now proceed to that box from right to left along the same row and after reaching to that particular box now go downwards column-wise.
3. Exit the Tableaux.
4. Count the number of boxes that you have came across or cut your way through.
5. That's the hook number for that particular box.
6. Calculate the hook number for all the boxes in the Tableaux.

For example $\begin{array}{|c|c|}\hline & \\ \hline \\ \cline{1-1} \end{array}$ has hook number $\begin{array}{|c|c|}\hline 3 & 1 \\ \hline 1 \\ \cline{1-1} \end{array}$. Where the value inside the box represents the value of the hook number of that box. Also this $\begin{array}{|c|c|c|}\hline & & \\ \hline & \\ \cline{1-2} \end{array}$ tableaux hook number is $\begin{array}{|c|c|c|}\hline 4 & 3 & 1 \\ \hline 2 & 1 \\ \cline{1-2} \end{array}$.

Now for a tableaux, you can calculate the dimensionality in SU(N) very easily.

3.2 Rules:

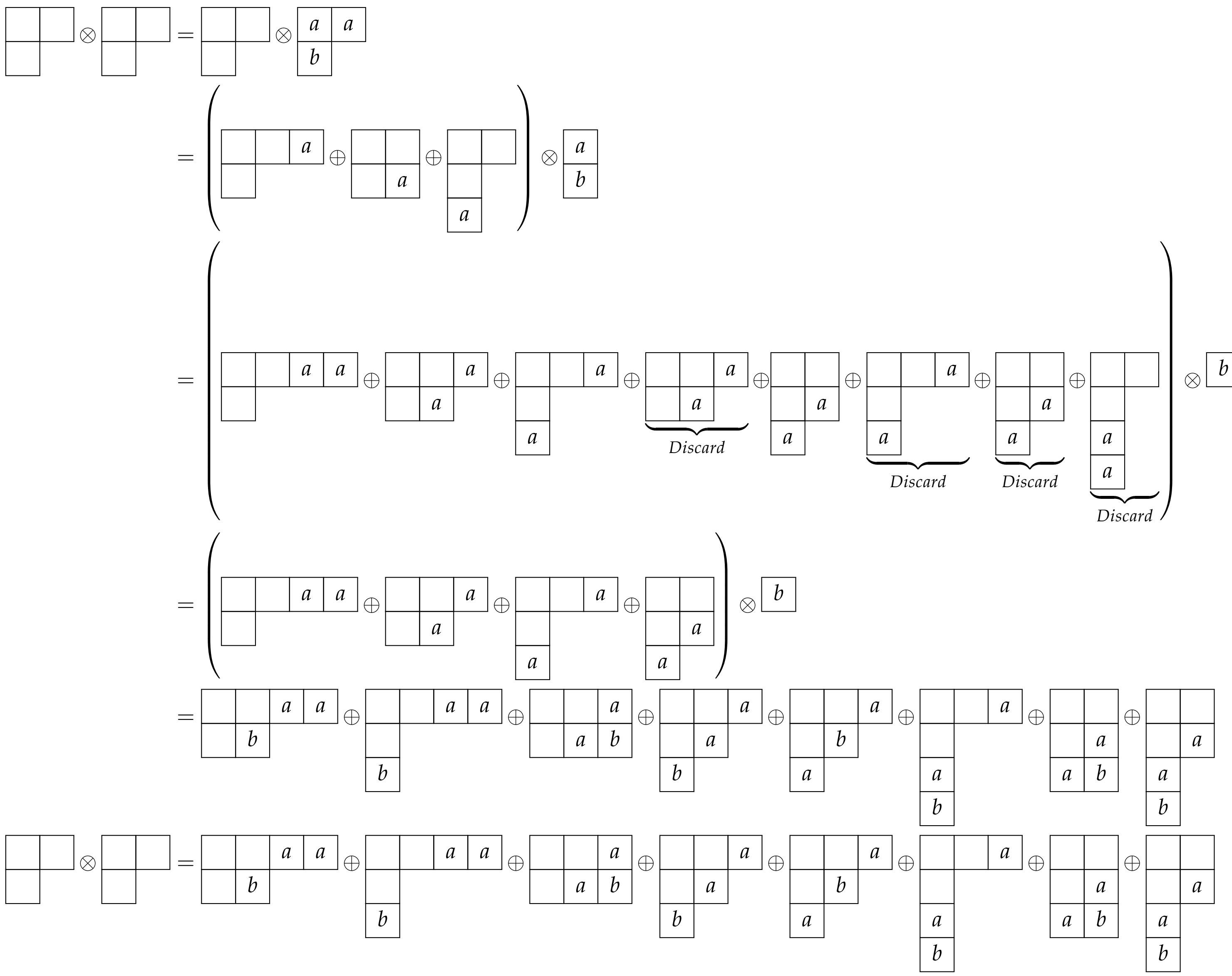
1. First go to very first row and find the box which is at very left.
2. Put the number N in that box. In the next box (same row but going right) Put $N+1$ and next one $N+2$ and so on.
3. Complete the first row, move on to next row.
4. Same procedure but in second row put the number $N-1$ in that very left box and in next box put N and next $N+1$ and so on.
5. Do the same for remaining rows.
6. While going left in a row increase the number and while going down decrease the number.
7. But the number of rows must not exceed the number N. In that case that tableaux is not valid in SU(N).
8. After putting all the numbers multiply them, say you get A.
9. Find the Hook number for each box and multiply them, say you get B.
10. The dimensionality of that Tableaux is $\frac{A}{B}$.

After filling up the numbers, the tableaux looks like this

3	2	1
2	1	

4	3	2	1
3	2	1	
2	1		

4 Tableaux Multiplication



$$(1,1)\otimes(1,1)=(2,2)\oplus(3,0,1)\oplus(0,3,0)\oplus(1,1,1)\oplus(1,1,1)\oplus(2,0,0,1)\oplus(0,0,3)\oplus(0,1,0,1)$$

In SU(3) the dimensionality is as follows:

$$\underbrace{(1,1)}_{8}\otimes\underbrace{(1,1)}_{8}=\underbrace{(2,2)}_{27}\oplus\underbrace{(3,0,1)}_{10}\oplus\underbrace{(0,3,0)}_{10}\oplus\underbrace{(1,1,1)}_{8}\oplus\underbrace{(1,1,1)}_{8}\ \underbrace{(2,0,0,1)}_{\text{Not exists in SU(3)}}\ \oplus\underbrace{(0,0,3)}_{1}\ \underbrace{(0,1,0,1)}_{\text{Not exists in SU(3)}} \tag{1}$$

so we have:

$$(8\otimes 8=27\oplus 10\oplus 10\oplus 8\oplus 8\oplus 1) \tag{2}$$

$$\ydiagram{2,1} \otimes \ydiagram{1,1} = \ydiagram{2,1} \otimes \begin{ytableau} a \\ b \end{ytableau}$$

$$= \left(\begin{ytableau} \ & \ & a \\ \ \end{ytableau} \oplus \begin{ytableau} \ & \ \\ \ & a \end{ytableau} \oplus \begin{ytableau} \ & \ \\ \ \\ a \end{ytableau} \right) \otimes \begin{ytableau} b \end{ytableau}$$

$$= \begin{ytableau} \ & \ & a & b \\ \ \end{ytableau} \oplus \begin{ytableau} \ & \ & a \\ \ & b \end{ytableau} \oplus \begin{ytableau} \ & \ & a \\ \ \\ b \end{ytableau} \oplus \begin{ytableau} \ & \ & b \\ \ & a \end{ytableau} \oplus \begin{ytableau} \ & \ \\ \ & a \\ b \end{ytableau} \oplus \begin{ytableau} \ & \ & b \\ \ \\ a \end{ytableau} \oplus \begin{ytableau} \ & \ \\ \ & b \\ a \end{ytableau} \oplus \begin{ytableau} \ & \ \\ \ \\ a \\ b \end{ytableau}$$

$$\ydiagram{2,1} \otimes \ydiagram{1,1} = \begin{ytableau} \ & \ & a \\ \ & b \end{ytableau} \oplus \begin{ytableau} \ & \ & a \\ \ \\ b \end{ytableau} \oplus \begin{ytableau} \ & \ \\ \ & a \\ b \end{ytableau} \oplus \begin{ytableau} \ & \ \\ \ \\ a \\ b \end{ytableau}$$

$(1,1) \otimes (0,1) = (1,2) \oplus (2,0,1) \oplus (0,1,1) \oplus (1,0,0,1)$

Now here are the set of rules:

- First assign same alphabet to each row. For example say I have 2 rows. So in 1st row each box I will write a, next row all boxes b and so on.

In SU(3) the dimensionality is as follows:

$$\underbrace{(1,1)}_{8} \otimes \underbrace{(0,1)}_{3} = \underbrace{(1,2)}_{15} \oplus \underbrace{(2,0,1)}_{6} \oplus \underbrace{(0,1,1)}_{3} \oplus \underbrace{(1,0,0,1)}_{\text{Not exists in SU(3)}} \tag{3}$$

so we have:

$$(8 \otimes 3 = 15 \oplus 6 \oplus 3) \tag{4}$$